



CAPSTONE PROJECT NOTES

HOUSE PRICE PREDICTION

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PROBLEM STATEMENT

Multiple recent events in the USA with domino effects have clearly demonstrated the need for studying and analyzing real estate trends. The real estate industry has been one of the leading researches focusing on modern economics, for its significant implications on relevant industries and fields such as construction, investment, and public welfare. In general, purchasing and investing in any real estate project will involve various transactions between different parties. Thus, it could be a vital decision for both households and enterprises.

1. Correlation between real estate prices and national economy: The financial crisis of 2008 has demonstrated a direct link between the ability of the public to afford their mortgage, and, the economic state of the country. This makes real estate prices an key indicator of underlying economic issues in the country, making data analysis in the field a valuable tool for the US Government in managing its economy

2. Decline in individual Home ownership: Even recently the New York times reported that 60 billion in middle class homes were lapped up by institutional buyers, pushing prices of these homes out of the reach of the common man. The Millennials and GenZ have been consistently vocal about not having the same opportunities of home ownership as the previous generations.

Understanding the elements that go into house pricing, and, analyzing their probable future value, gives us a baseline. Although housing prices will continue to fluctuate over the years, if we have a base line we can adjust to the growth and inflation to understand if all is going well or if there is some unexpected turns in the market.

Thus building a model to help us to predict the house prices, in this case of Washington state, is a step in the right direction towards saving the market from another catastrophe. This model works to:

- To identify the variables affecting house prices, e.g. area, number of rooms, bathrooms, etc.
- To create a linear model that quantitatively relates house prices with variables such as number of rooms, area, number of bathrooms, etc.
- To know the accuracy of the model, i.e. how well these variables can predict house prices.



The Data we have has 21613 rows and 23 columns. The information has been collected until the year 2015 and there is not particular indication that it was phased collection. So we can assume it was all collected in on go in the year 2015.

The list features can be seen below:

1. cid: a notation for a house
2. dayhours: Date house was sold
3. price: Price is prediction target
4. room_bed: Number of Bedrooms/House
5. room_bath: Number of bathrooms/bedrooms
6. living_measure: square footage of the home
7. lot_measure: square footage of the lot
8. ceil: Total floors (levels) in house
9. coast: House which has a view to a waterfront
10. sight: Has been viewed
11. condition: How good the condition is (Overall)
12. quality: grade given to the housing unit, based on grading system
13. ceil_measure: square footage of house apart from basement
14. basement_measure: square footage of the basement
15. yr_built: Built Year
16. yr_renovated: Year when house was renovated
17. zipcode: zip
18. lat: Latitude coordinate
19. long: Longitude coordinate
20. living_measure15: Living room area in 2015(implies-- some renovations) This might or might not have affected the lotsize area
21. lot_measure15: lotSize area in 2015(implies-- some renovations)
22. furnished: Based on the quality of room
23. total_area: Measure of both living and lot

A sample of the data is given below:

	cid	dayhours	price	room_bed	room_bath	living_measure	lot_measure	ceil	coast	sight	...	basement	yr_built	yr_renovated	zipcode
0	3878100940	20150427T000000	800000	4.0	1.75	3050.0	9440.0	1	0	0.0	...	1250.0	1988	0	98034
1	3145800250	20150317T000000	190000	2.0	1.00	670.0	3101.0	1	0	0.0	...	0.0	1948	0	98118
2	7129303070	20140820T000000	735000	4.0	2.75	3040.0	2415.0	2	1	4.0	...	0.0	1988	0	98118
3	7338220280	20141010T000000	257000	3.0	2.50	1740.0	3721.0	2	0	0.0	...	0.0	2009	0	98002
4	7950300670	20150218T000000	450000	2.0	1.00	1120.0	4590.0	1	0	0.0	...	0.0	1924	0	98118

measure	lot_measure	ceil	coast	sight	...	basement	yr_built	yr_renovated	zipcode	lat	long	living_measure15	lot_measure15	furnished	total_area
3050.0	9440.0	1	0	0.0	...	1250.0	1988	0	98034	47.7228	-122.183	2020.0	8860.0	0.0	12490
670.0	3101.0	1	0	0.0	...	0.0	1948	0	98118	47.5548	-122.274	1680.0	4100.0	0.0	3771
3040.0	2415.0	2	1	4.0	...	0.0	1988	0	98118	47.5188	-122.256	2620.0	2433.0	0.0	5455
1740.0	3721.0	2	0	0.0	...	0.0	2009	0	98002	47.3383	-122.213	2030.0	3794.0	0.0	5461
1120.0	4590.0	1	0	0.0	...	0.0	1924	0	98118	47.5883	-122.285	1120.0	5100.0	0.0	5710



Further analyzing the features:

	count	mean	std	min	25%	50%	75%	max
cid	21613.000000	4580301520.864988	2876565571.312048	1000102.000000	2123049194.000000	3904930410.000000	7308900445.000000	9900000190.000000
price	21613.000000	540182.158793	367362.231718	75000.000000	321950.000000	450000.000000	645000.000000	7700000.000000
room_bed	21505.000000	3.371355	0.930289	0.000000	3.000000	3.000000	4.000000	33.000000
room_bath	21505.000000	2.115171	0.770248	0.000000	1.750000	2.250000	2.500000	8.000000
living_measure	21596.000000	2079.880761	918.498121	290.000000	1429.250000	1910.000000	2550.000000	13540.000000
lot_measure	21571.000000	15104.583283	41423.619385	520.000000	5040.000000	7618.000000	10684.500000	1651359.000000
sight	21556.000000	0.234366	0.766438	0.000000	0.000000	0.000000	0.000000	4.000000
quality	21612.000000	7.656857	1.175484	1.000000	7.000000	7.000000	8.000000	13.000000
ceil_measure	21612.000000	1788.366556	828.102535	290.000000	1190.000000	1560.000000	2210.000000	9410.000000
basement	21612.000000	291.522534	442.580840	0.000000	0.000000	0.000000	560.000000	4820.000000
yr_renovated	21613.000000	84.402258	401.679240	0.000000	0.000000	0.000000	0.000000	2015.000000
zipcode	21613.000000	98077.939805	53.505026	98001.000000	98033.000000	98065.000000	98118.000000	98199.000000
lat	21613.000000	47.560053	0.138564	47.155900	47.471000	47.571800	47.678000	47.777600
living_measure15	21447.000000	1987.065557	685.519629	399.000000	1490.000000	1840.000000	2360.000000	6210.000000
lot_measure15	21584.000000	12766.543180	27286.987107	651.000000	5100.000000	7620.000000	10087.000000	871200.000000
furnished	21584.000000	0.196720	0.397528	0.000000	0.000000	0.000000	0.000000	1.000000

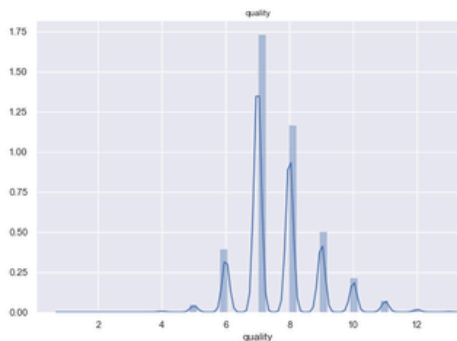
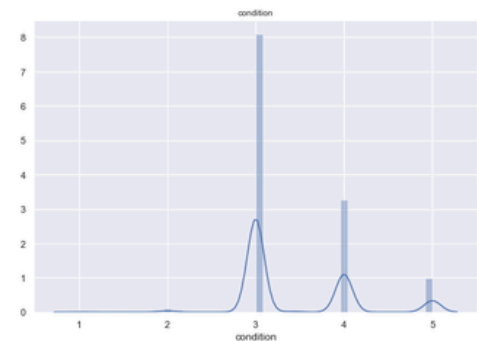
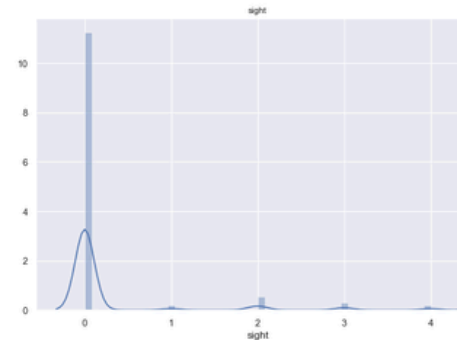
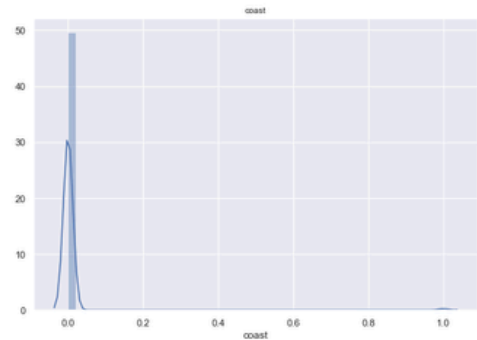
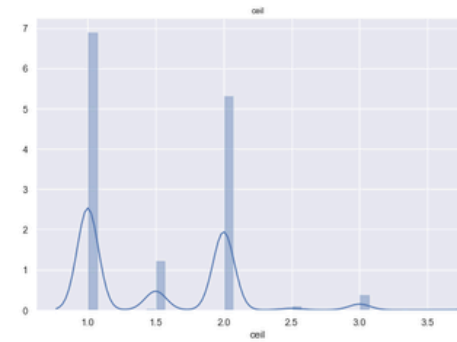
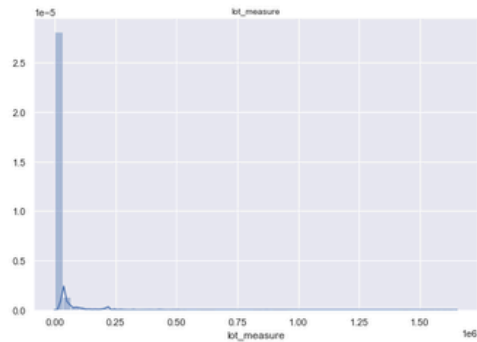
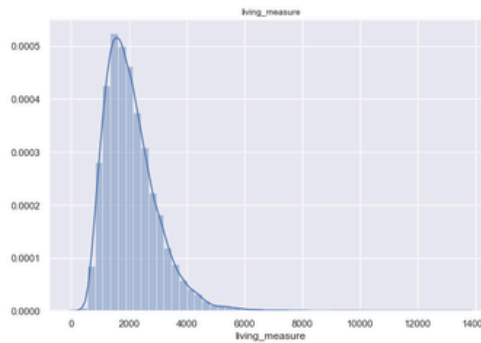
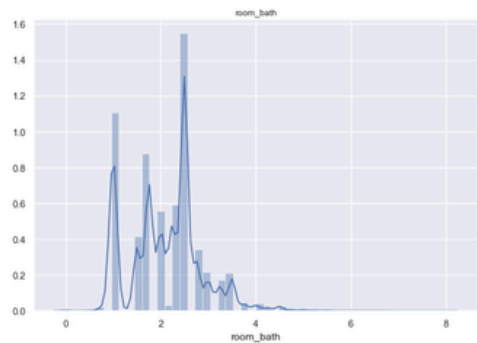
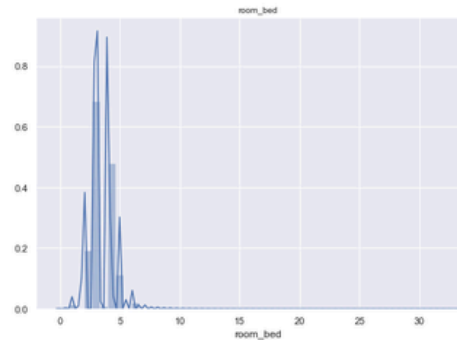
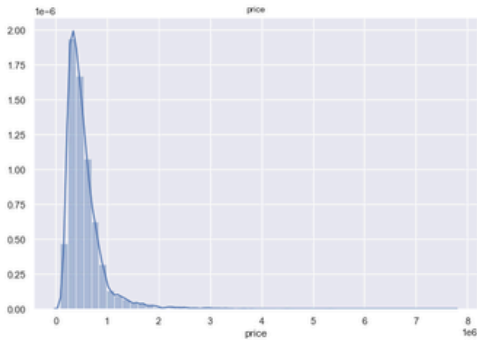
We observe that there are some missing values in the features right away. Also we note that for a few fields the max value is further away from the 75% of the feature. This could indicate outliers.

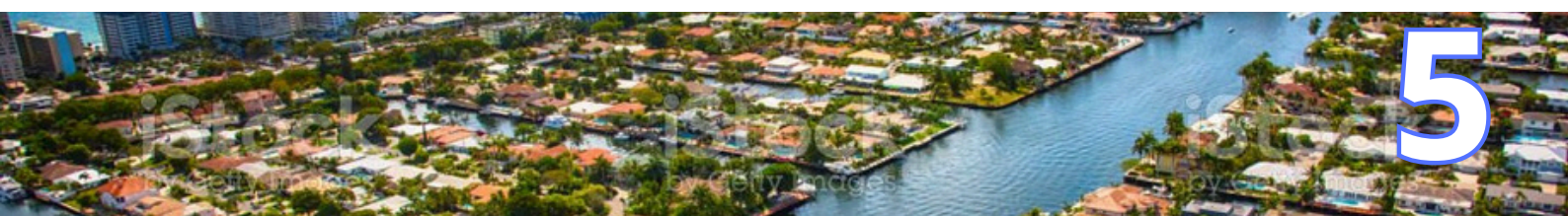
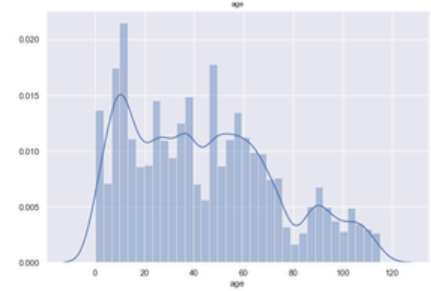
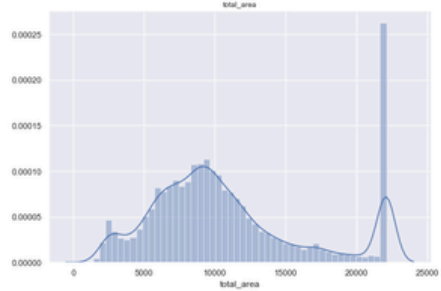
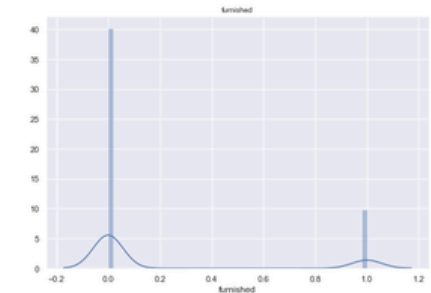
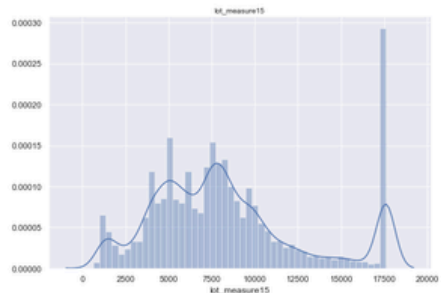
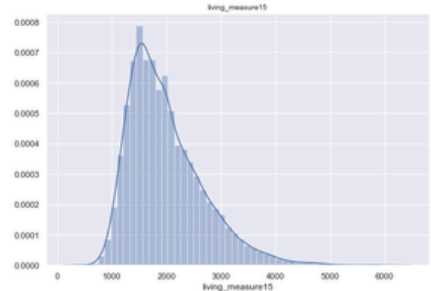
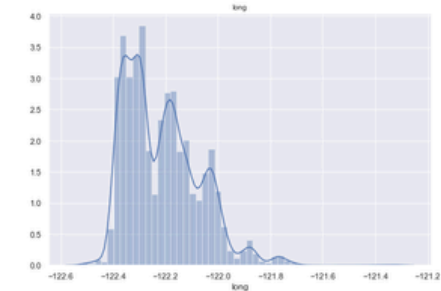
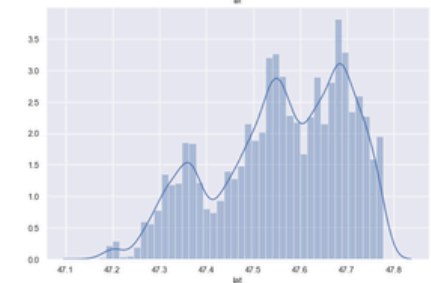
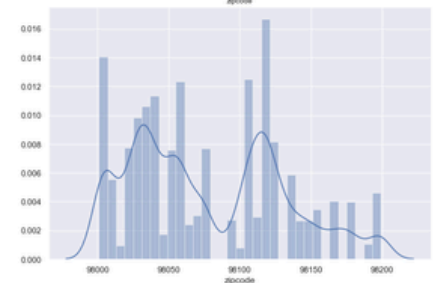
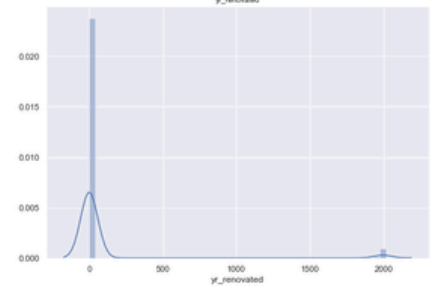
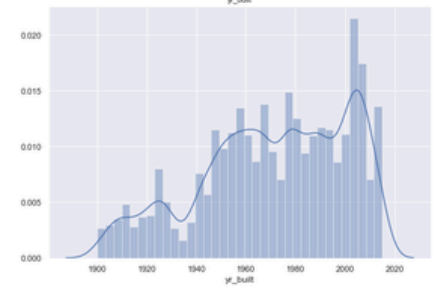
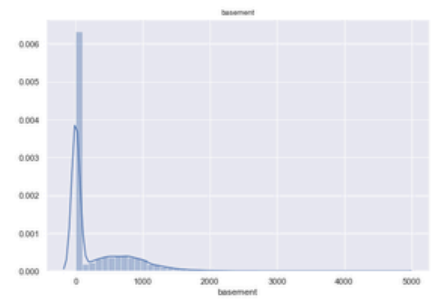
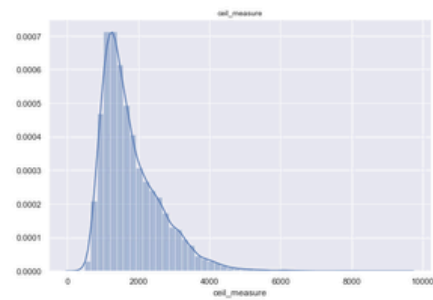
We will next clean the data and make it more appropriate for our model building process.



EXPLORATORY DATA ANALYSIS

UNIVARIATE ANALYSIS



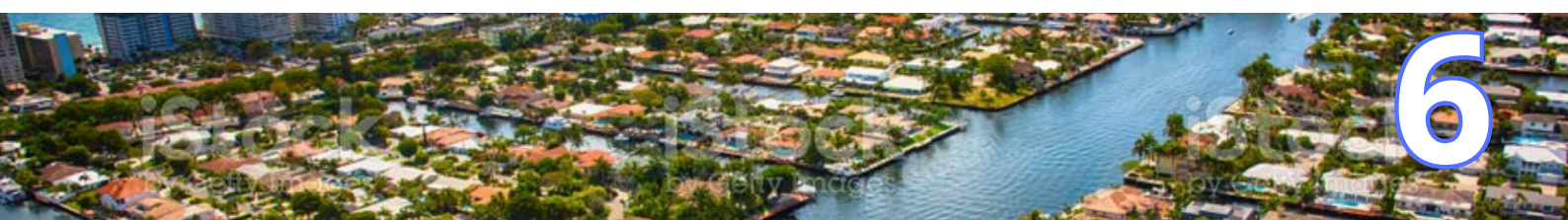


From the above distribution plots we see that our target variable has outliers which is making it left skewed.

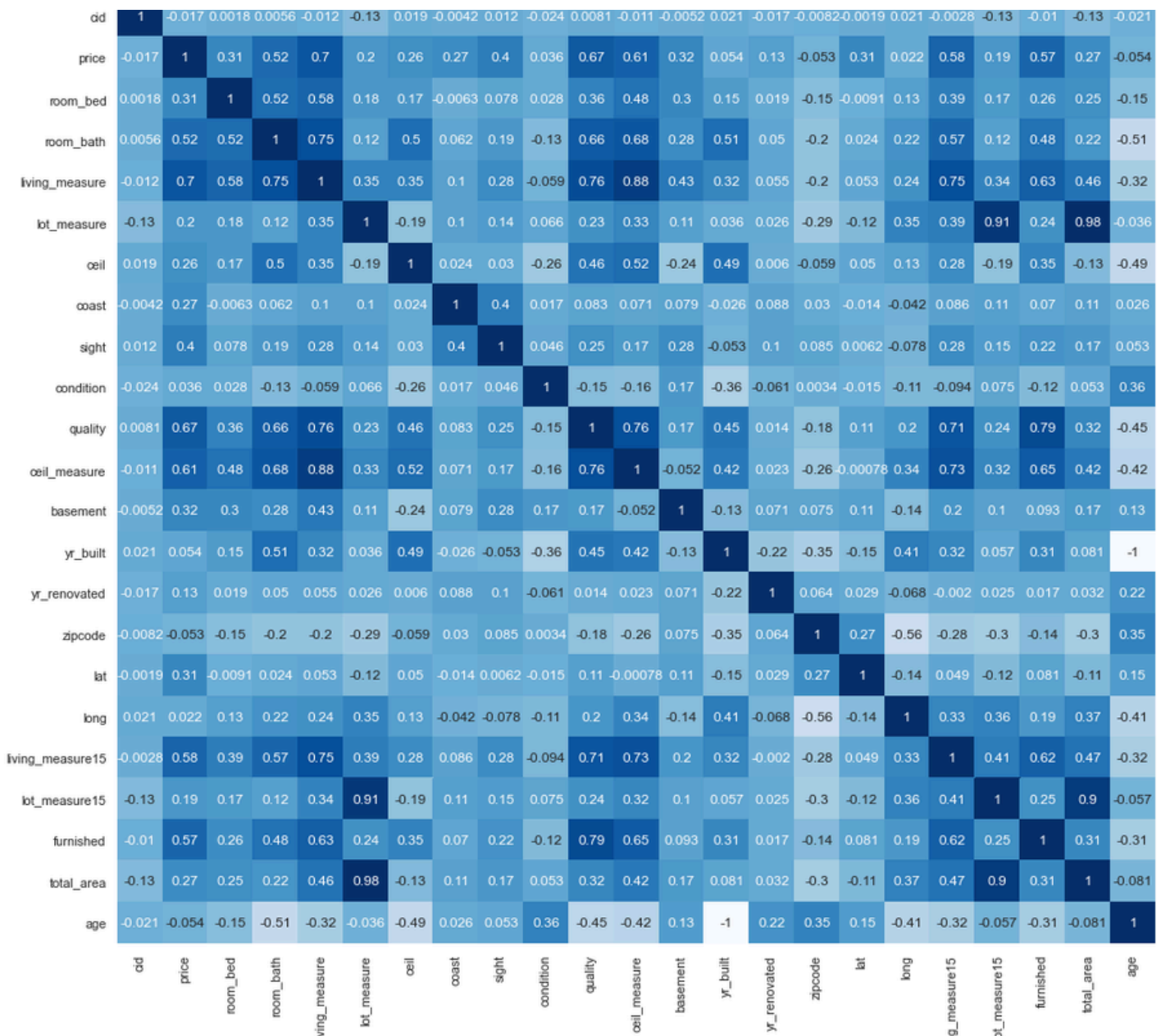
Other than that we can also see that most of the continuous variables are also left skewed in distribution. Outlier treatment might rectify the case and help with our model performance.

Variable `Sight` contains most records of house not being viewed. May be majority of the properties under consideration are not for sale.

The Variable `Furnished` shows there are very few furnished properties on the market.



CORRELATION ANALYSIS

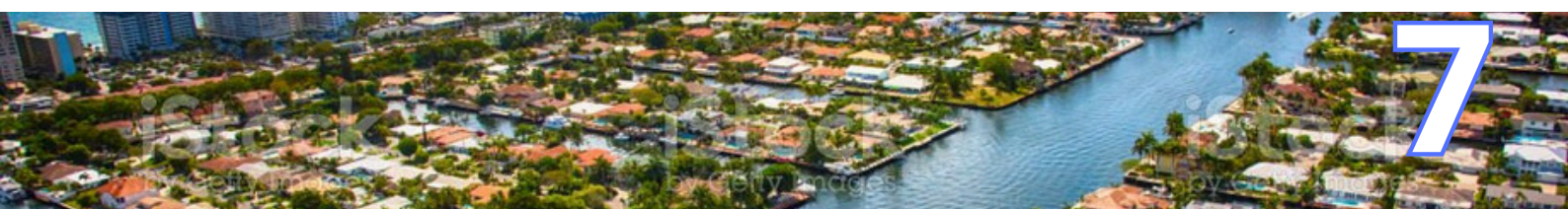


The Correlation maps shows which features are highly correlated with the price of houses. we can see that features Sight Quality and Furnished have high values.

There is also high correlation between various measures of the property like Lot measure and Living Measure, Ceiling measure and Living Measure. these might just be indicative of building requirements.

Features Furnished and quality are also highly correlated.

The new feature Age has significant correlation with Zip code, Quality, Ceiling measure, Condition and number of bath rooms.



CLUSTERING ANALYSIS

Row Labels	Count of cid	Min of price	Max of price	Average of price
0	4553	84000	1990000	493217.3707
1	8180	75000	1440000	397310.6709
2	5184	139500	1850000	464469.7718
3	3535	257000	7700000	990927.6611
4	161	285000	7060000	1668300.882
Grand Total	21613	75000	7700000	540182.1588

In an attempt to understand if there are any patterns in the data that we can readily take advantage of I have clustered the data using K-Means clustering.

But from the results there is no obvious group on the Price.

Low and High value properties seem to be distributed across the segments and are not making the Price baskets apparent.

But it might help the model to understand the correlation of the field to predict the price.



MODEL BUILDING APPROACH

PCA

- Multiple Correlated columns
- Helps reduce dimensions and helps feed better data to model

Base model building

- Linear Regression
- Sets a baseline for the data.

Exploration of other models

- Models Chosen: KNN, Random forest, Gradient Boost
- Observe which works best for the data.

Model evaluation

- Evaluate Score, MAPE and RMSE across models for both original and Post-PCA Data.
- Compare the scores to understand where we have good performance and consistency of results across test and train.



MODEL BUILDING

Before the Model Building process I have considered PCA to combine the features of the house in the hope of reducing some of the correlation.

I have reduced the feature from 16 to 8 and will be running my set of models on both the Original data set and the post-PCA data set.

Based on the result we can understand if PCA was indeed useful in the current scenario.

Post PCA I Have choose to build the following models:

- Multi linear regression - The Linear regressor is built as a base model to set a baseline of the data/model performance.
- KNN regressor - Following that the KNN regressor will evaluated the if we can use nearest neighbor to predict he house price. As i have performed clustering based on this algorithm I would like to see if this model is able to give good results on the data.
- Random Forest Regressor - Next we will use the Random Forest regressor. The Random Forest algorithm is formed from multiple decision trees that are constructed in a random way. This includes randomly sampling our data and randomly selecting variables/features from our dataset for each tree. This I expect will be a more robust approach and has the potential to give a good result.
- Gradient Boost Regressor - Finally in an attempt to further tune the results I will be implementing a Gradient boost model.

I have attached the Model Score for each of the above in comparison with the PCA data as well:

Model Name	Original Data		Original Data		Original Data	
	Train Score	Test Score	Train MAPE	Test MAPE	Train RMSE	Test RMSE
Linear Regression	0.6558	0.6619	4.4916	4.6125	0.5868	0.5809
KNN	0.7816	0.7081	3.2313	3.8208	0.4673	0.5398
Random Forest	0.7355	0.7237	3.4264	3.6103	0.5143	0.5252
Gradient Boost	0.7937	0.7577	3.5559	3.7022	0.4542	0.4918

Model Name	Post PCA Data		Post PCA Data		Post PCA Data	
	Train Score	Test Score	Train MAPE	Test MAPE	Train RMSE	Test RMSE
Linear Regression	0.6078	0.6204	4.8305	5.0361	0.6263	0.6156
KNN	0.7703	0.6933	2.9309	3.9323	0.4793	0.5533
Random Forest	0.7051	0.6832	3.6744	3.6011	0.5431	0.5618
Gradient Boost	0.7431	0.6931	3.8046	3.7203	0.5069	0.5535



MODEL EVALUATION

From the above table we understand that there is no model that stands out as a best fit. All the performance scores are in the same range and hence we will have to choose the best among these for our purposes.

I have the following observations:

- The Model performances for the original data and the Post PCA data are not that far off.
- Hence I am able to conclude that PCA has not improved the data presented to the models.
- The KNN Model has improved from the base model but is not the best option.
- The random forest model has great results on the train data but the performance on the test suggest that it might be over fitting on the data. Hence this model might not be the best suited to predict future data.
- The gradient boost model gives consistent performance across train and test and has the highest score among the models.
- The first attempt at the model yielded a decent result but there was still room for refining.
- Post the tuning we have arrived at the results previously displayed.

Model Description	Data Source	Score	MAPE	RMSE
Original Data	Train	0.8234	3.3889	0.4203
	Test	0.7597	3.6459	0.4898
PCA Data	Train	0.7942	3.8046	0.5069
	Test	0.7002	3.7012	0.5471

Though the values are not as high as might be expected in the industry, this model is best suited for the current data set and will be useful in predicting house prices.



RECOMMENDATIONS

- From the Bivariate and PCA we understand that the dimensions that affect the price the most are:
- Age, Condition, Coast, Sight and Number of Bedrooms
- Spacious apartments with good views are clearly most in demand and hence the most expensive.
- New projects should consider maximising the space of the units.
- High rises offering good views will also be valued higher than other apartments.
- A floor charge can be levied by builders/promoters to take advantage of the customers wanting good views.
- Customers looking to fit properties within a budget can look for older properties.
- They can also consider the Lower floors of big complexes. This will give them access to the amenities at a lower price range.

